

Design of Closed-Loop In-Plane Suspended Gate FET (CLIP-SGFET) based accelerometer

Ganga K M

Roll No: EE21MS010

Supervisor: Prof. Naveen Kadayinti

Co-supervisor : Prof. Seena V



॥ सा विद्या या विमुक्तये ॥
ಭ್ರ.ತಂ.ಸಂ. ಧಾರವಾಡ
भा. प्रौ. सं. धारवाड
I.I.T. DHARWAD

Department of Electrical, Electronics and Communication Engineering
INDIAN INSTITUTE OF TECHNOLOGY DHARWAD

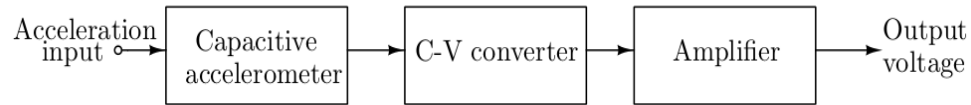
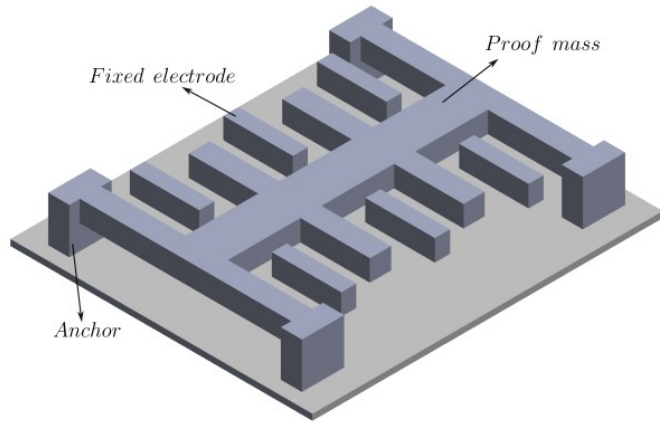
July 11, 2024

Overview

- 1) Introduction
- 2) Modeling of devices and design of circuits involved in CLIP-SGFET based accelerometer
- 3) Design improvements in CLIP-SGFET based accelerometer
- 4) Response to reviewer's comments
- 5) Conclusions and Future work

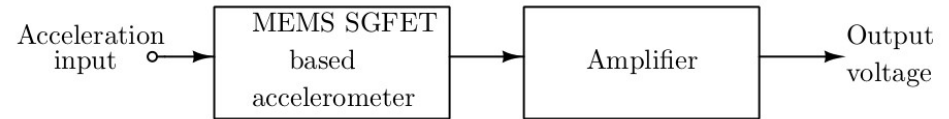
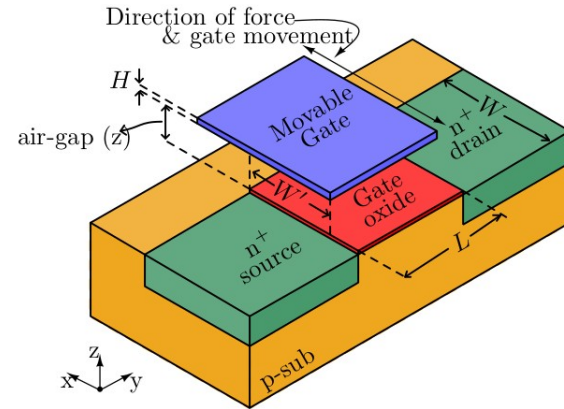
Introduction

Conventional capacitive accelerometers



- Consumes lot of area
- Low sensitivity
- Extra circuitry : C-V converter

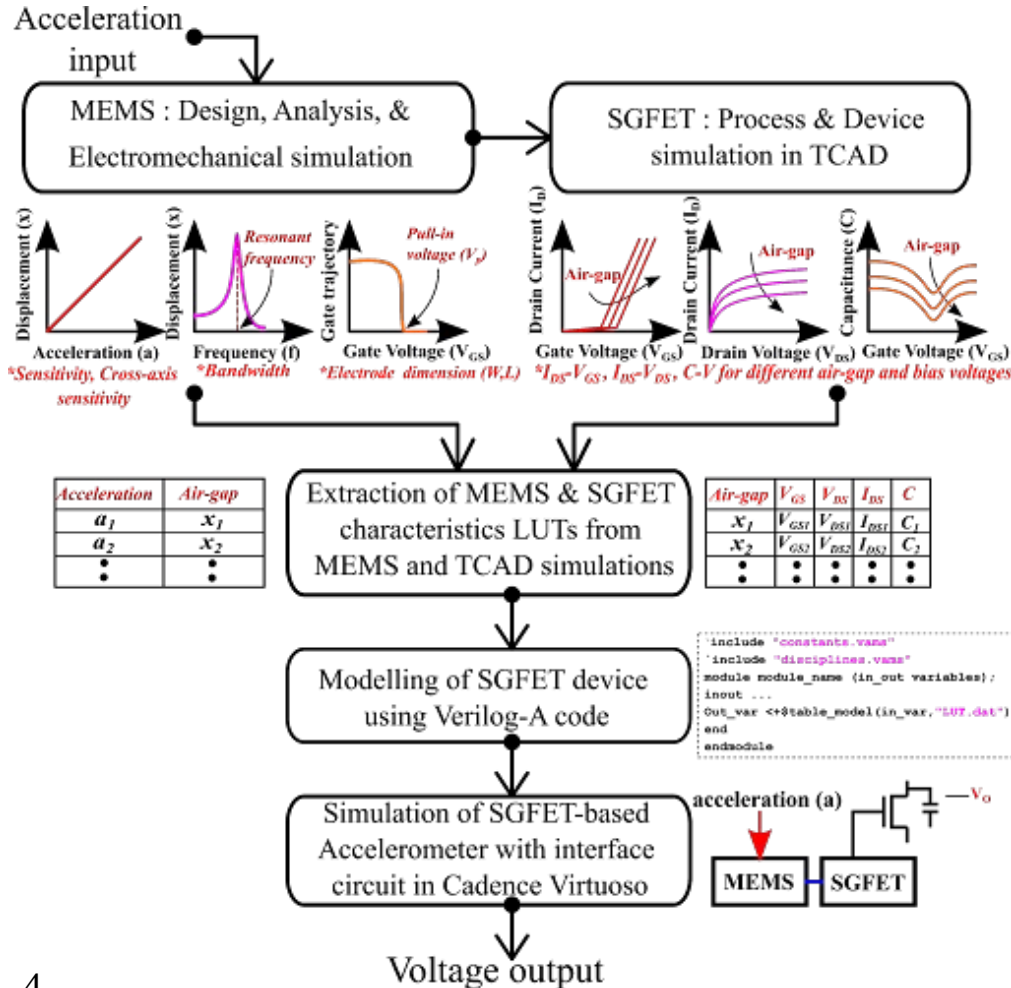
SGFET-based accelerometers



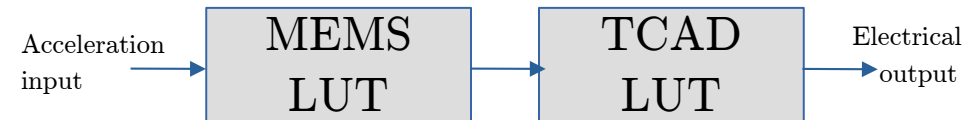
- Direct transduction
- Lesser area
- Higher sensitivity

Modeling of devices
and
Design of circuits
involved in CLIP-SGFET based accelerometer

Modeling the devices : LUT method

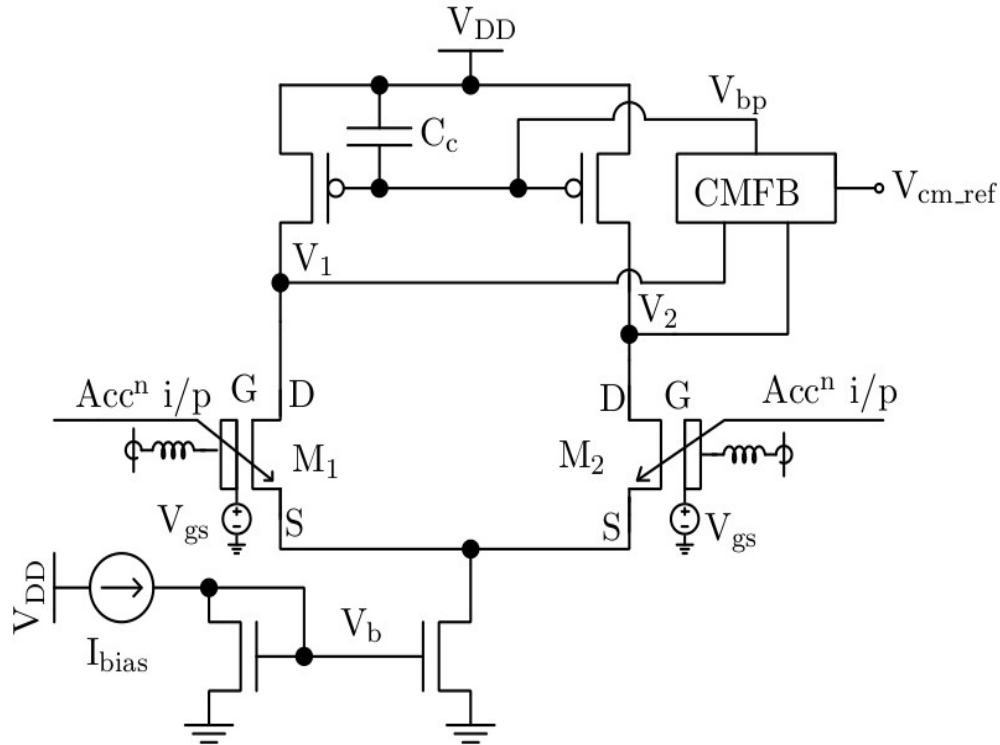


- All possible widths are obtained through electromechanical simulation in MEMS+
- Device simulations performed for different terminal voltages for these set of widths and the final current is obtained through TCAD simulation.



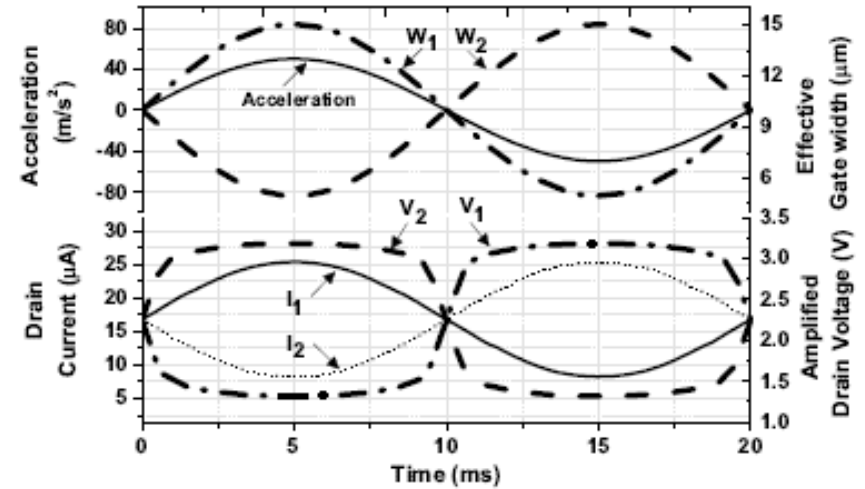
- These are encoded as voltages in Verilog-A models, which is used in Cadence simulations.

Open loop accelerometer



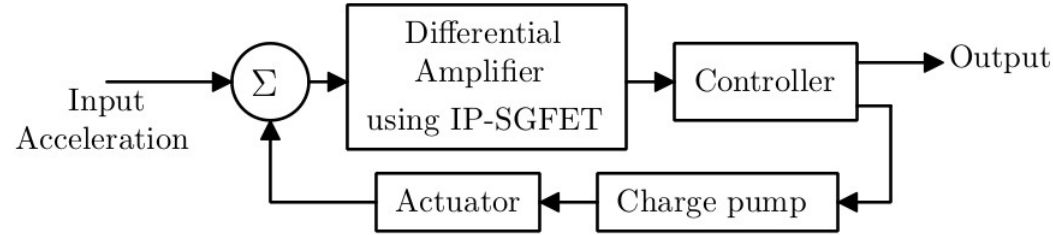
- SGFET : LUT model is used.
- V_{DD} : 3.3V, I_{bias} : 6uA, V_{gs} : 2V

Transient simulation



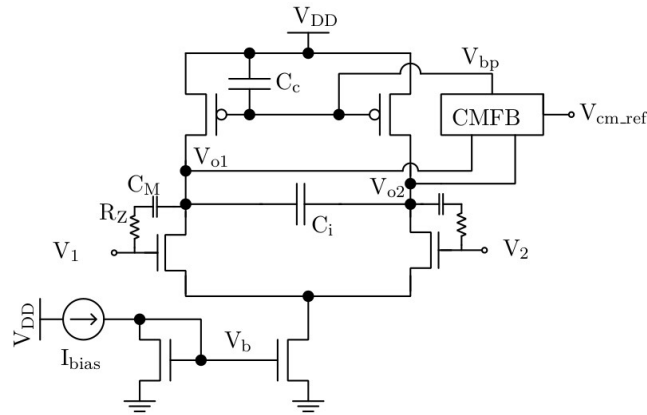
- Output voltage saturates due to high gain.
- Small signal gain : 927 mV/g
- Closed-loop operation necessary to obtain high dynamic range.

Closed loop accelerometer

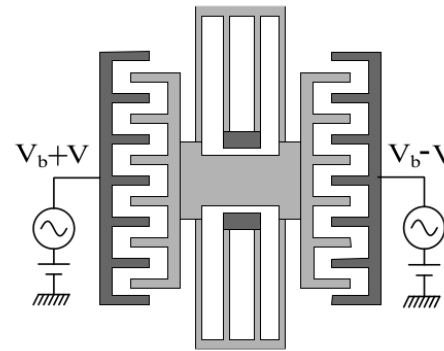


With differential amplifier(IP-SGFETs as input pair), closed loop consists of :

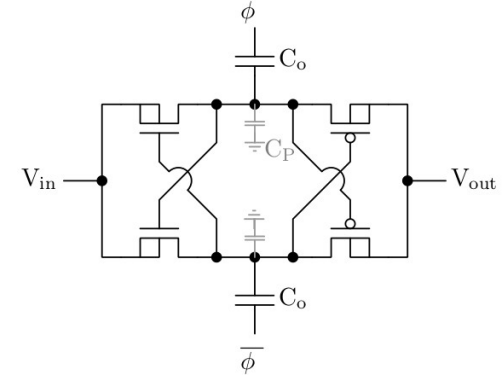
1. Controller



2. Actuator

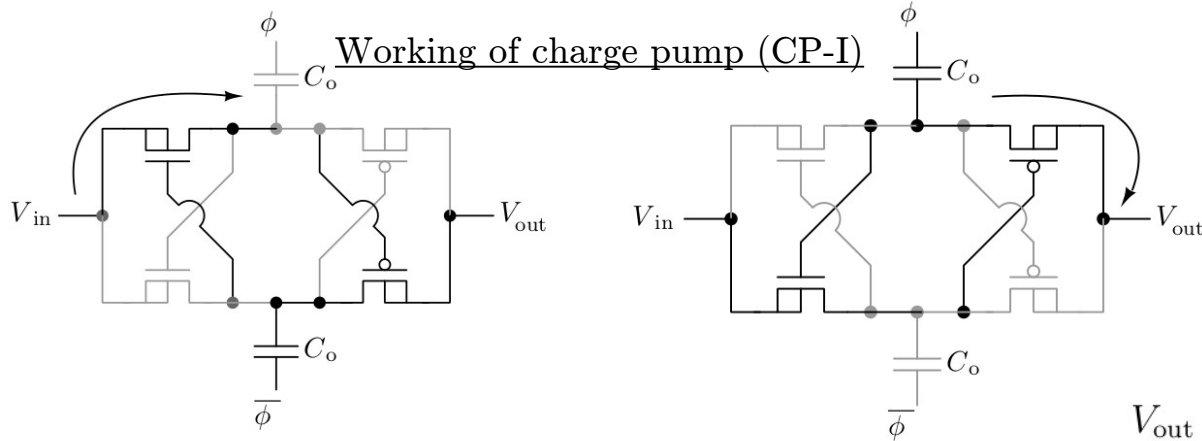


3. Charge pump



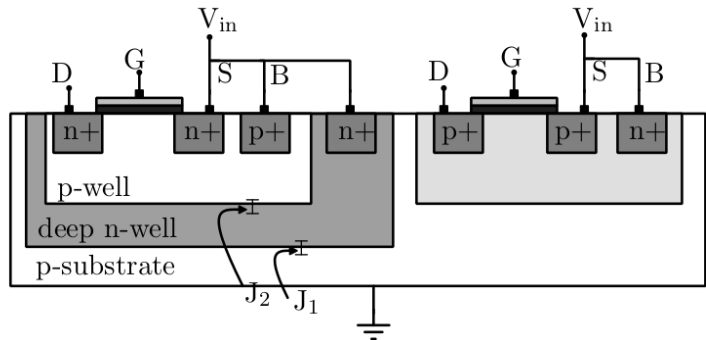
Data extracted from MEMS+ simulations and modeled as LUT using Verilog-A is used Cadence simulations.

Charge pumps for high voltage generation

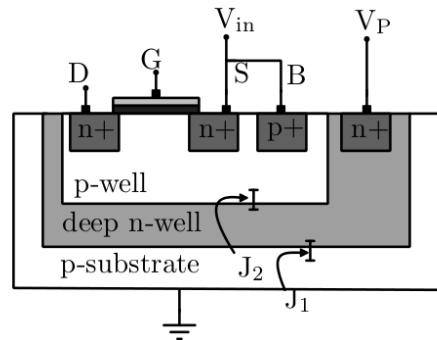


(a) Phase I: Voltage transfer mode
(clock = low)

(b) Phase II: Voltage pumping mode
(clock = high)



(a) CP-I



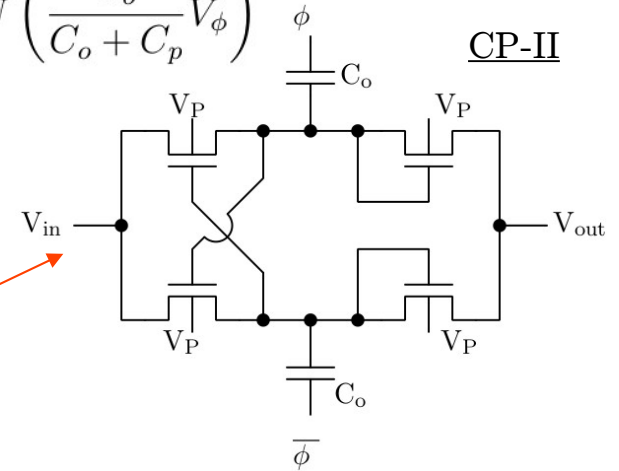
(b) CP-II

- Uses triple-well transistors.
- There is current flow from input to output. No reverse path.

What is the maximum voltage that can be generated?

- Limitations of CP-I : reverse breakdown voltage of the n-well/p-substrate junction (J_1)

$$V_{out} = V_{in} + N \left(\frac{C_o}{C_o + C_p} V_{\phi} \right)$$

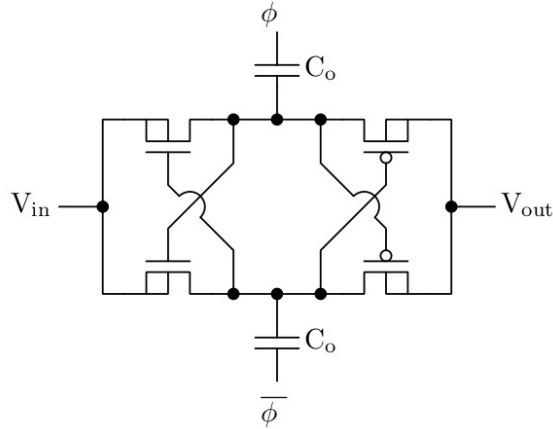


Need for reconfigurability

- Multiple stages of CP-I and CP-II cascaded to generate higher voltage.

Requirements :

- The high voltage generated should be continuous and proportional to input acceleration.
- Path for discharge when input reduces.



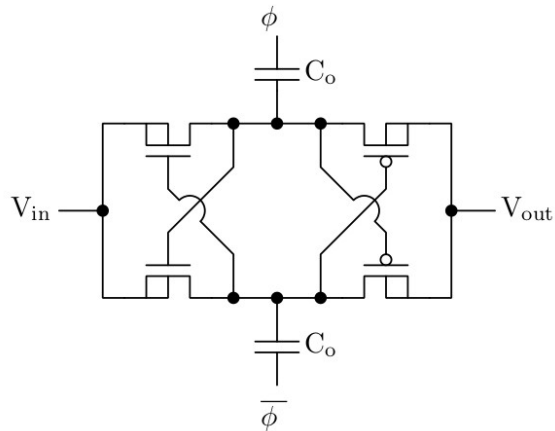
$$V_{\text{out}} = V_{\text{in}} + N \left(\frac{C_o}{C_o + C_p} V_{\phi} \right)$$

V_{out} can be controlled by :

- Varying number of active stages, N
- Varying efficiency, C_o

Need for reconfigurability

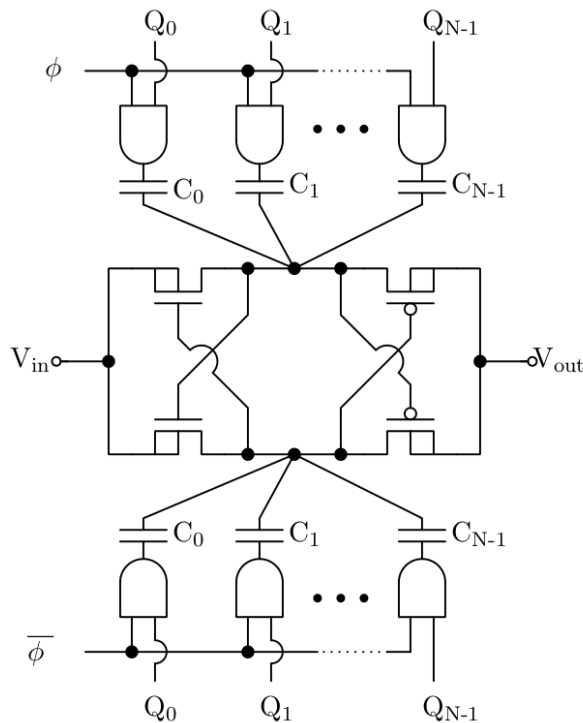
- Multiple stages of CP-I and CP-II cascaded to generate higher voltage.
- The voltage generated should be continuous and proportional to input acceleration.



$$V_{\text{out}} = V_{\text{in}} + N \left(\frac{C_o}{C_o + C_p} V_{\phi} \right)$$

V_{out} can be controlled by :

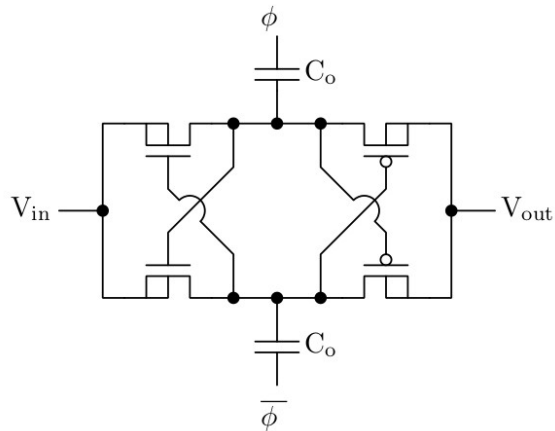
- Varying number of active stages, N
- Varying efficiency, C_o



- Varying efficiency by controlling the number of capacitors coupled with clock.
- Suits our latency requirements.

Need for reconfigurability

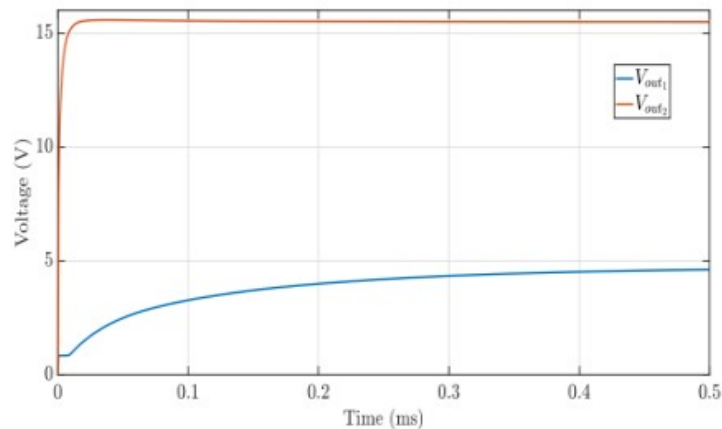
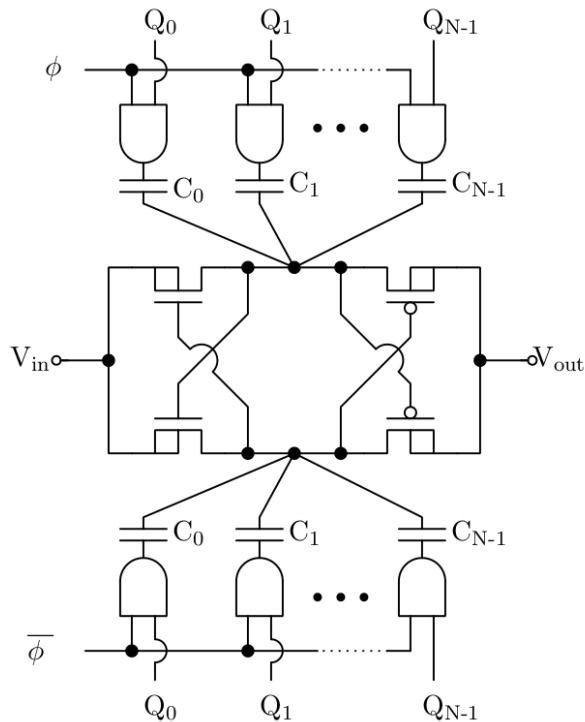
- Multiple stages of CP-I and CP-II cascaded to generate higher voltage.
- The voltage generated should be continuous and proportional to input acceleration.



$$V_{out} = V_{in} + N \left(\frac{C_o}{C_o + C_p} V_{\phi} \right)$$

V_{out} can be controlled by :

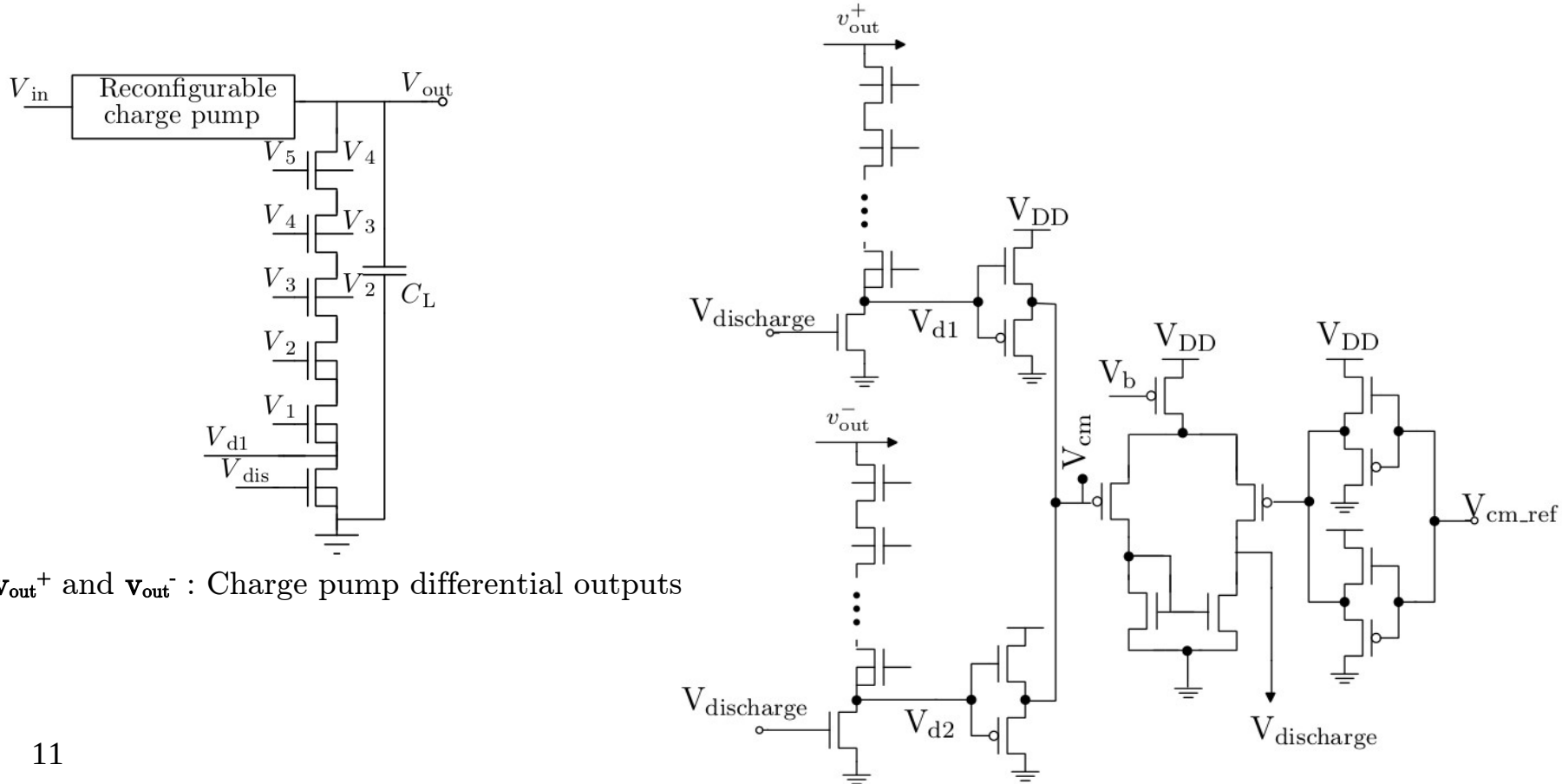
- Varying number of active stages, N
- Varying efficiency, C_o



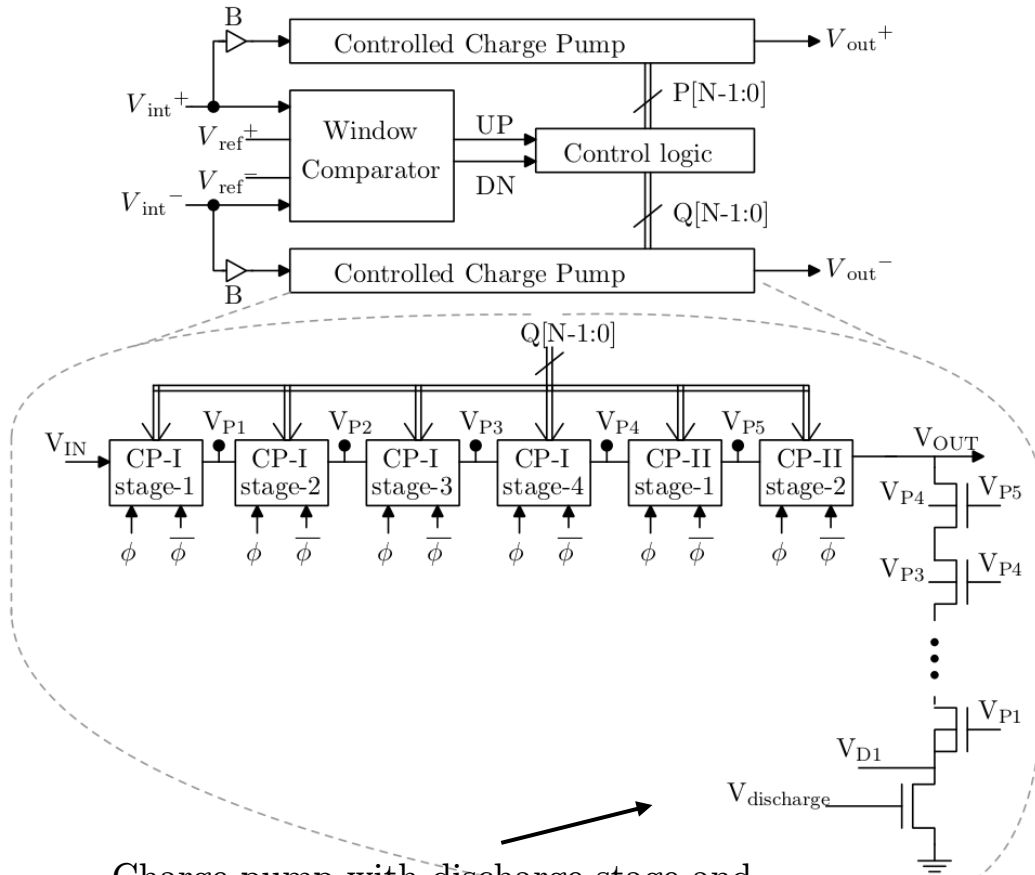
- Varying efficiency by controlling the number of capacitors coupled with clock.
- Suits our latency requirements.

Discharge stage

- Charge pumps don't allow reverse current flow.
- Discharge stage facilitates discharge of output node when input reduces.

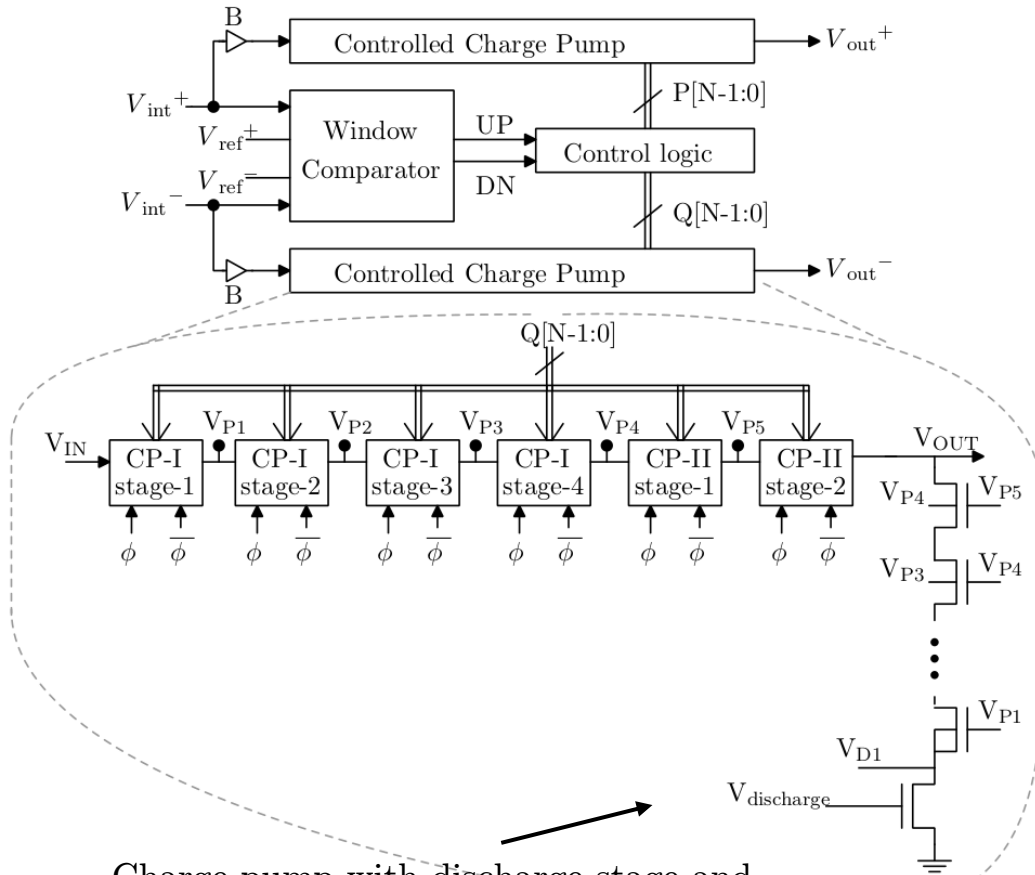


Complete schematic of closed loop

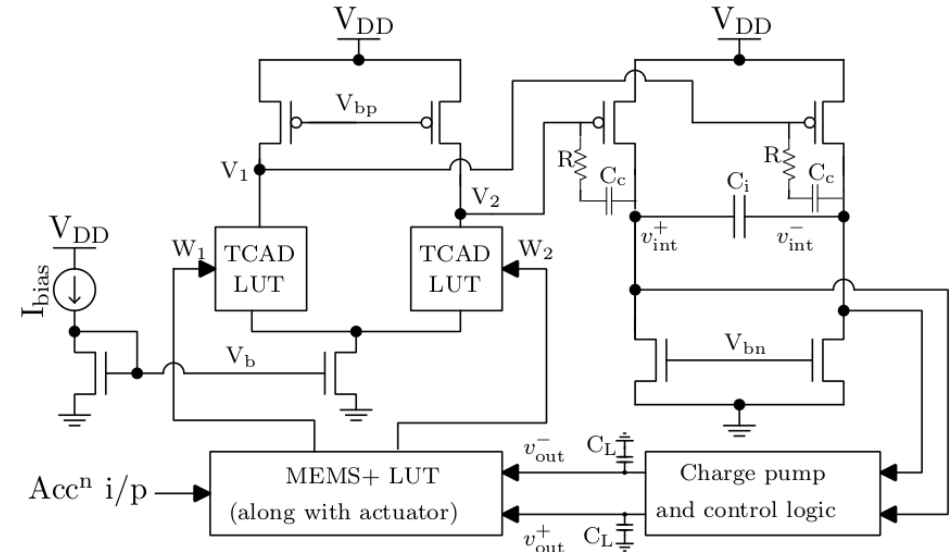


Charge pump with discharge stage and digital control logic which controls the charge pump efficiency

Complete schematic of closed loop

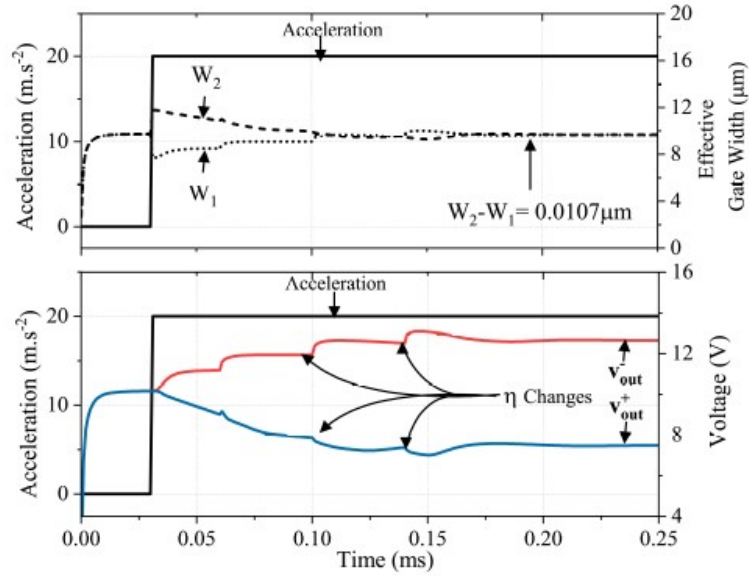


Charge pump with discharge stage and digital control logic which controls the charge pump efficiency

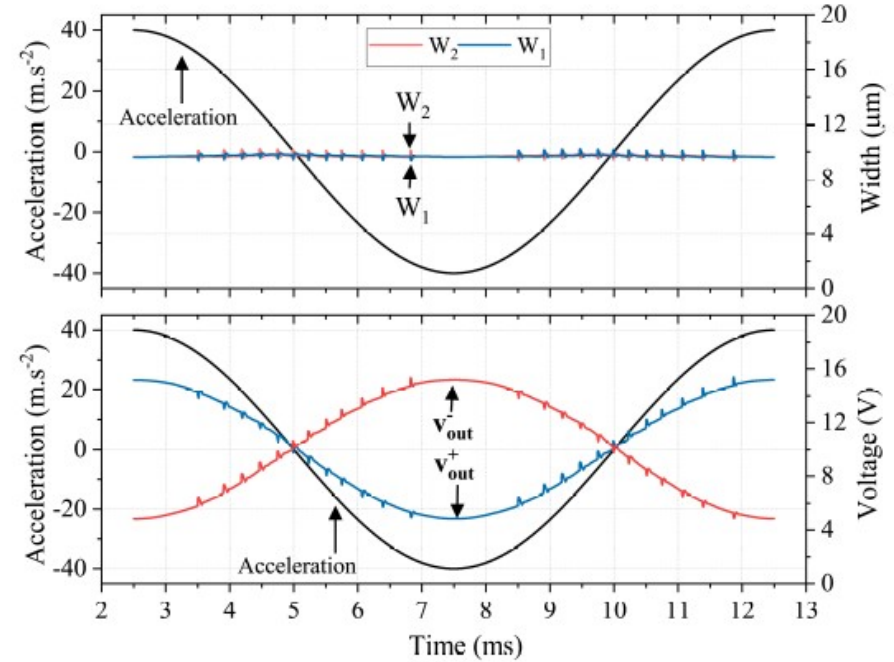


Complete schematic of closed-loop system

Simulated results



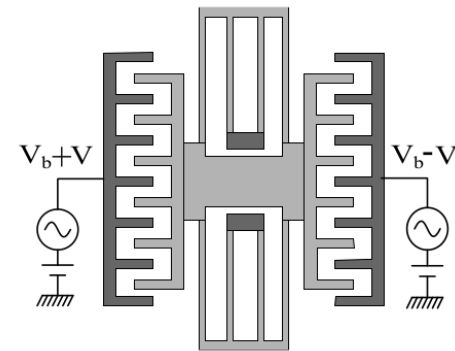
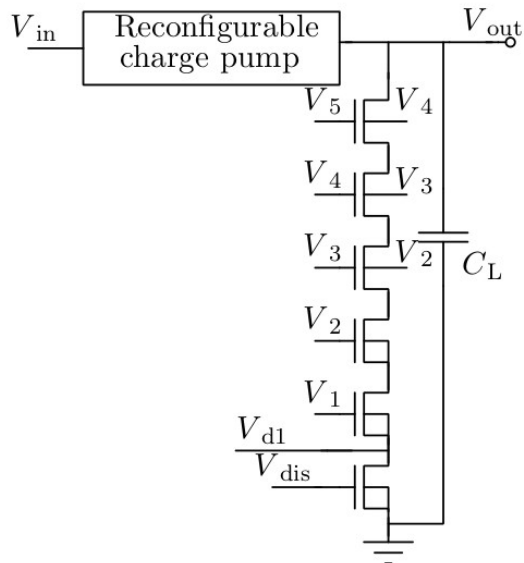
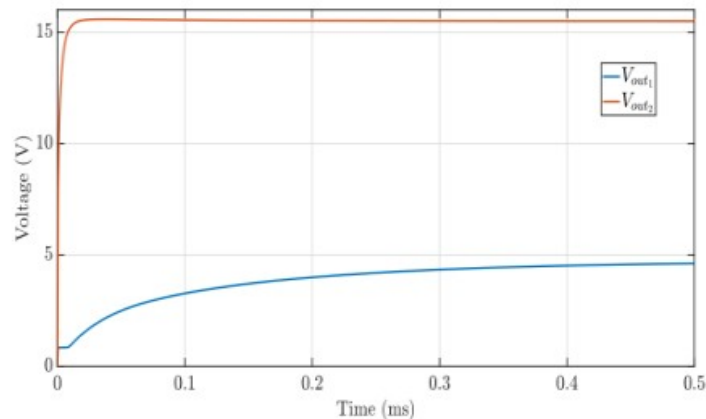
Step response for input step of 2g at 25us



Transient response for sine wave input of range $\pm 4g$ at frequency 100Hz

Design improvements in CLIP-SGFET based accelerometer

Design improvements in closed-loop

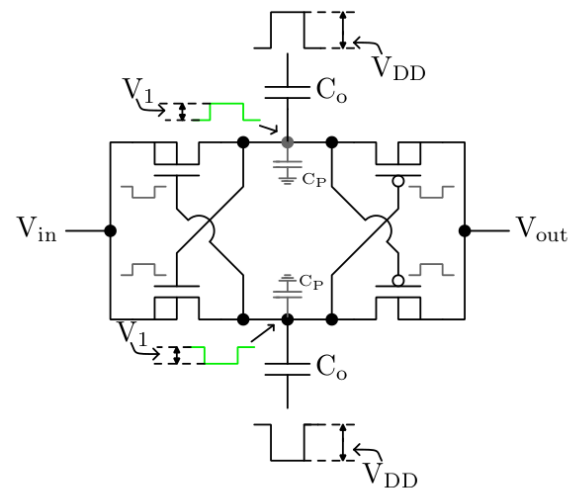


- The variations in the latency varies over a wide range from 430 ns to 70us.
- Reducing the variations improves system performance.
- Existing LUT does not capture frequency dependent behaviour.
- Effective resistance of the discharge path is not constant as the gate voltages vary.
- Inaccurate common-mode voltage measurement.

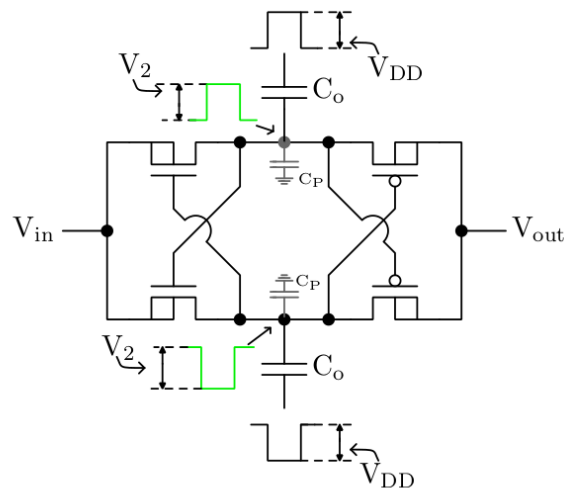
Latency variations in charge pump

- The variations in the latency still vary over a wide range from 430 ns to 70us.
- Reducing the variations reduces design complexity.

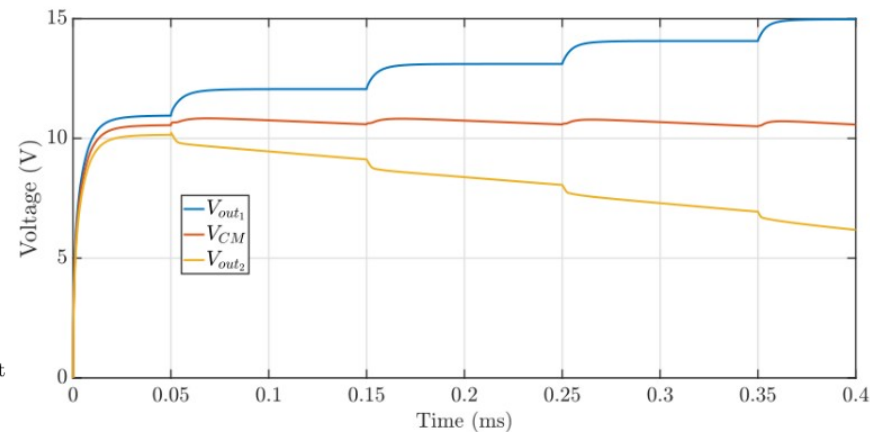
Why do these variations occur?



(a) At least efficiency, coupling capacitor, voltage added by each stage of CP-I is $V_1 = 0.5$ V. Here, $C_o = 250$ fF, Efficiency $= \eta = 15.15\%$.



(b) At highest efficiency, coupling capacitor, voltage added by each stage of CP-I is $V_2 = 2.3$ V. Here, $C_o = 1.45$ pF, Efficiency $= \eta = 69.69\%$.

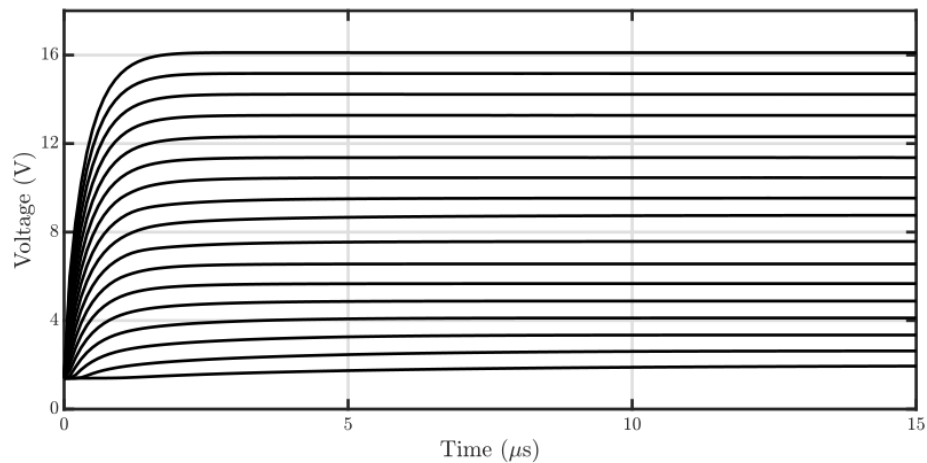
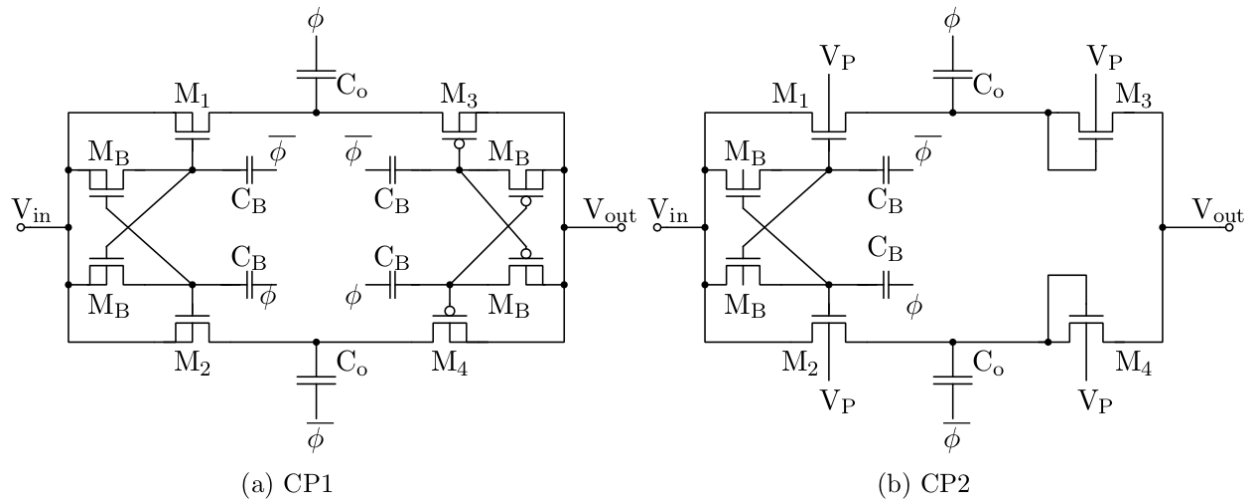


Variations in measured common-mode voltage.

$$R_{DS(on)} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

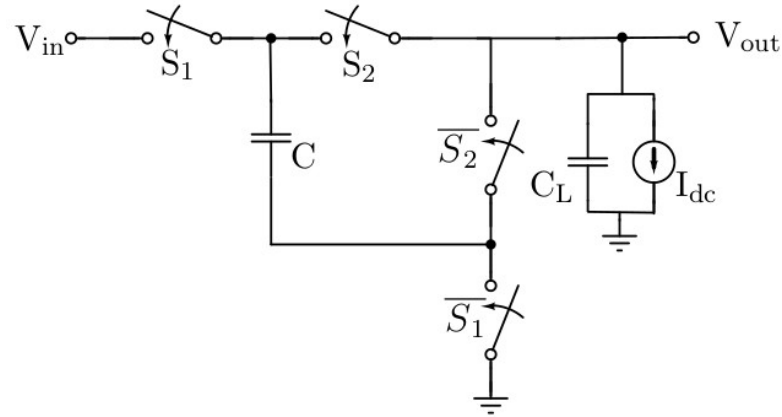
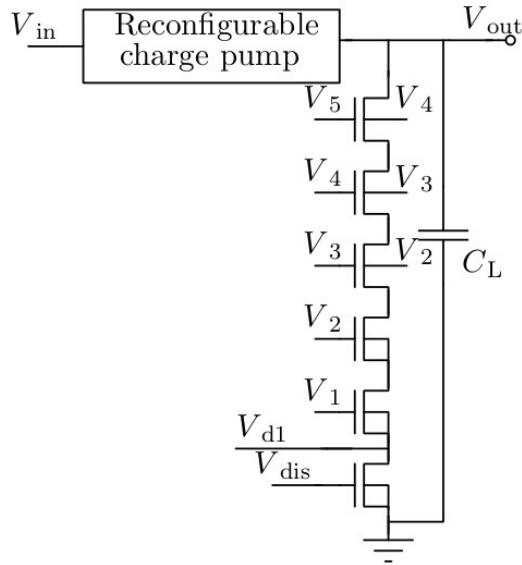
Latency variations in charge pump

Bootstrapped switches



← Output voltage at various efficiency settings for charge pump with bootstrapped switches

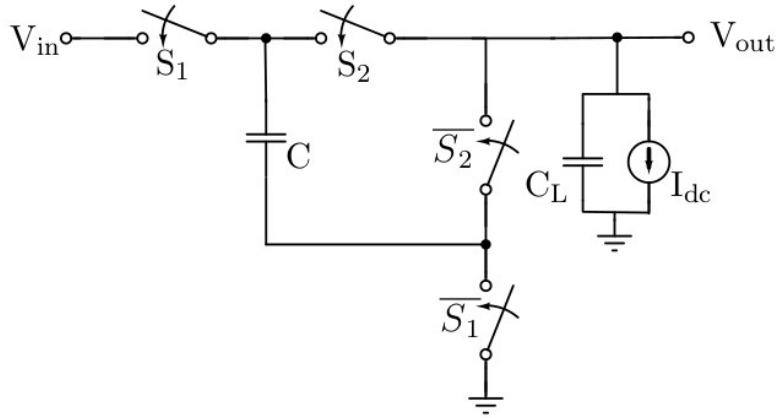
Multi-level converters as discharge stage



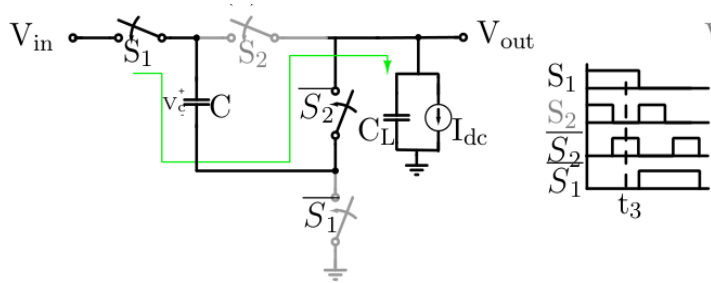
- Multi-level converters are DC-DC converter circuits.
- With appropriate switching sequence, the output voltage can be set to required level between 0V and input voltage level.
- The output voltage depends on the switch state at that instant.

- Effective resistance of the discharge path is not constant as the gate voltages vary.
- Inaccurate common-mode voltage measurement.

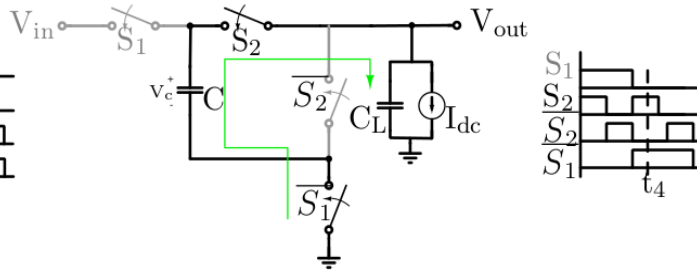
Multi-level converters as discharge stage



- Multi-level converters are DC-DC converter circuits.
- With appropriate switching sequence, the output voltage can be set to required level between 0V and input voltage level.
- The output voltage depends on the switch state at that instant.



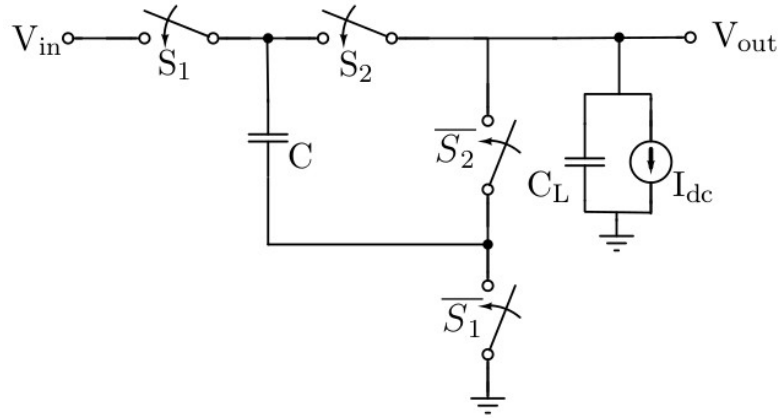
$$V_{out} = V_{in} - V_c$$



$$V_{out} = V_c$$

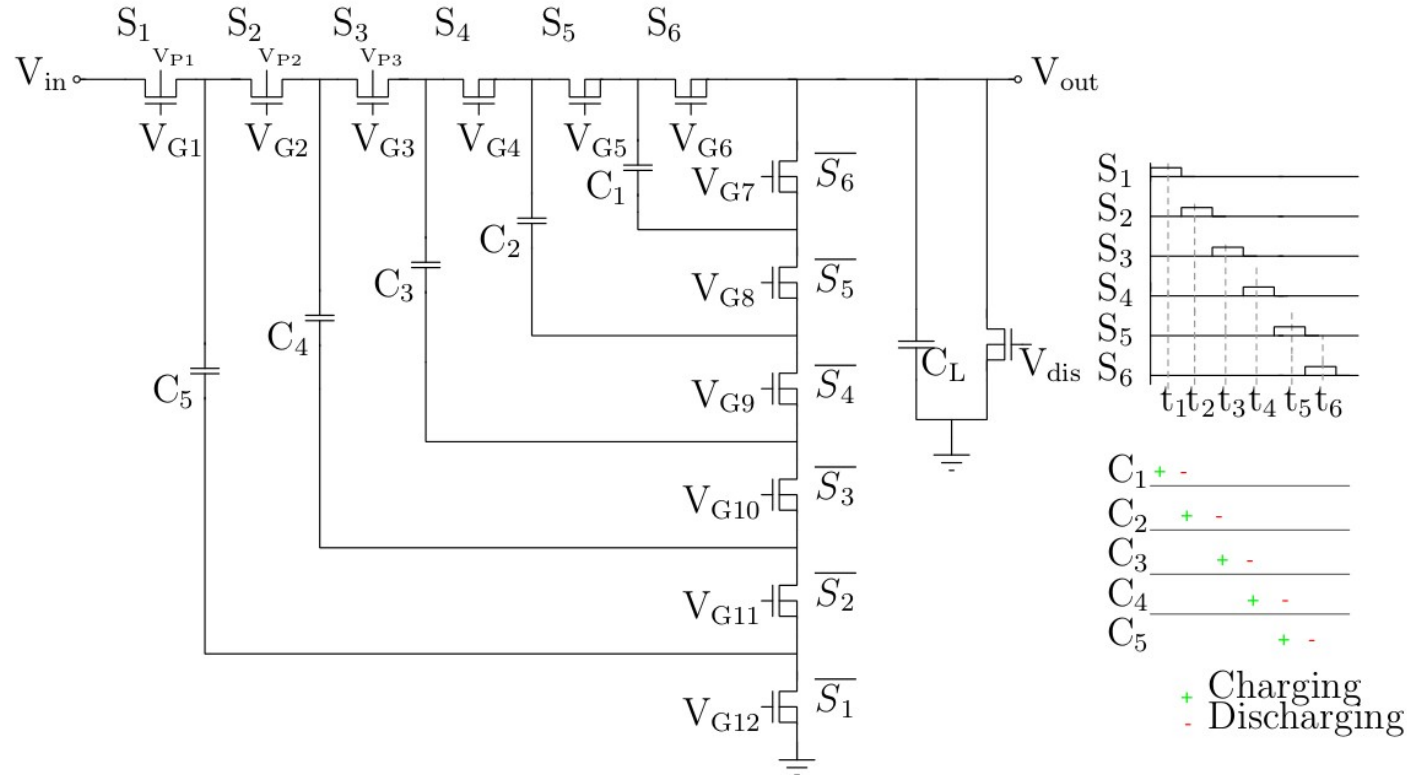
Multi-level converters as discharge stage

Requirements : Step down the output of the charge pump (which goes up to 17 V) by maintaining the right voltage difference across the terminals of all the switches without leading to a breakdown.

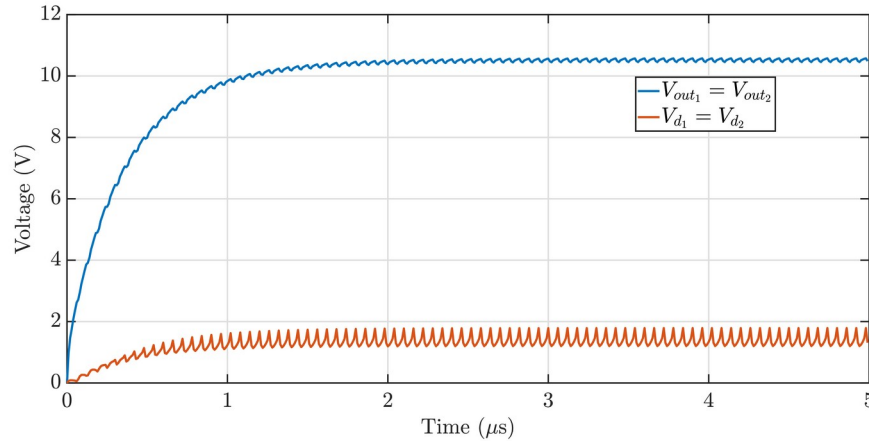


Multi-level converters as discharge stage

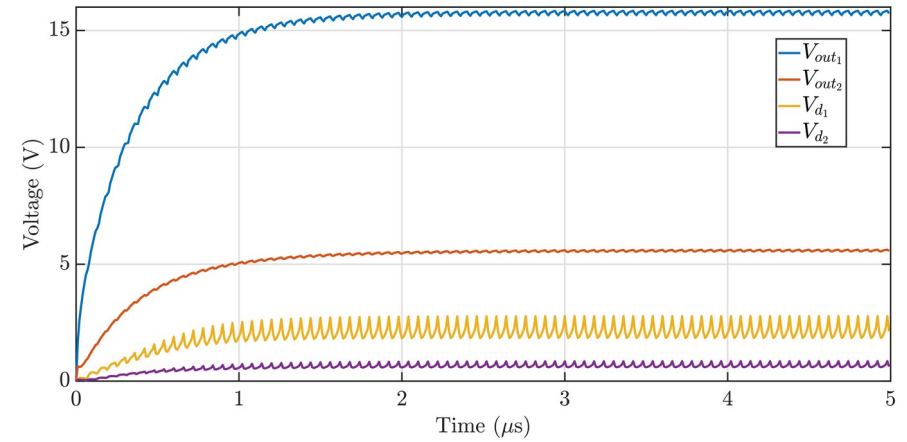
Requirements : Step down the output of the charge pump (which goes up to 17 V) by maintaining the right voltage difference across the terminals of all the switches without leading to a breakdown.



Multi-level converters as discharge stage



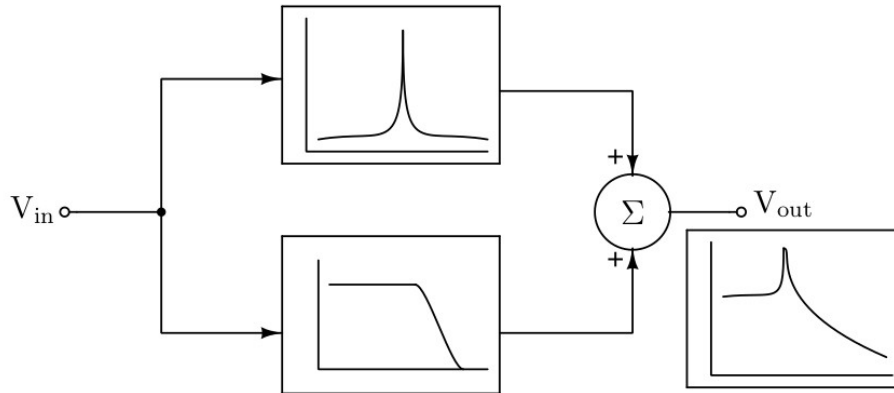
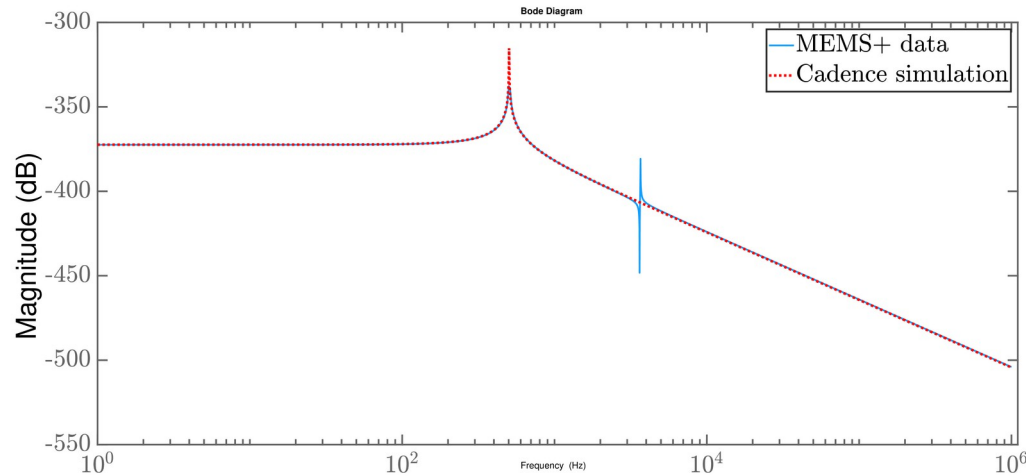
(a) $V_{out1}=V_{out2}=10.5\text{V}$, $V_{d1}=V_{d2}=1.4\text{V}$



(b) $V_{out1}=15.77\text{V}$, $V_{out2}=5.58\text{V}$, $V_{d1}=2.16\text{V}$ $V_{d2}=0.67\text{V}$

Simulated results of digitally reconfigurable charge pump at (a) common-mode (b) differential efficiency setting.

AC analysis of actuator



- AC analysis of the actuator structure is done in MEMS+ for a DC operating point of 10.5 V.
- A circuit with the same transfer function of the actuator structure is designed.

AC response of actuator has a peak at 505Hz(resonant frequency of the strcuture).

- We connect oscillator in parallel with 2nd order LPF to match the response is the plot shown.

Post layout simulation results

Layout

- Design was submitted to 2023 UNIC-CASS program
- **Title** : Design of Analog Front end (AFE) circuits for Closed-Loop IP-SGFET based accelerometer
- Accepted for sponsored tapeout

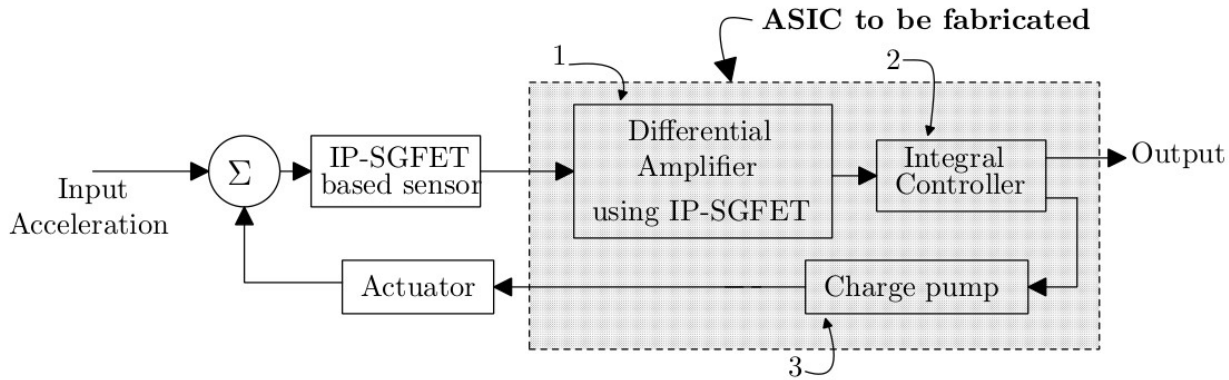
Tools used in designing :

1 . Analog flow :

- Xschem(schematic capture and post layout simulations)
- Magic(layout and DRC)

2 . Digital flow :

- OpenLane(RTL to GDS)

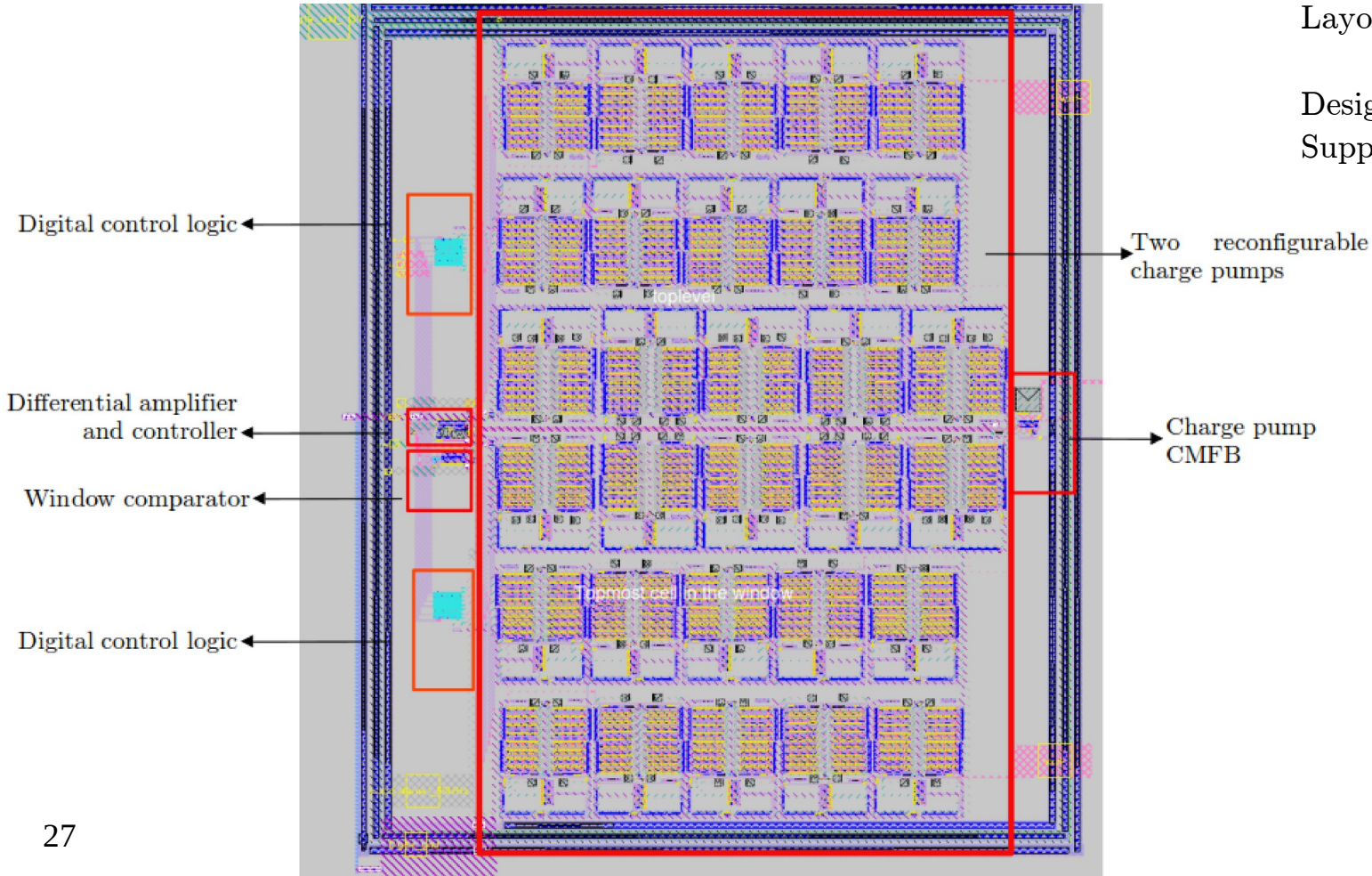


Complete layout of accelerometer system

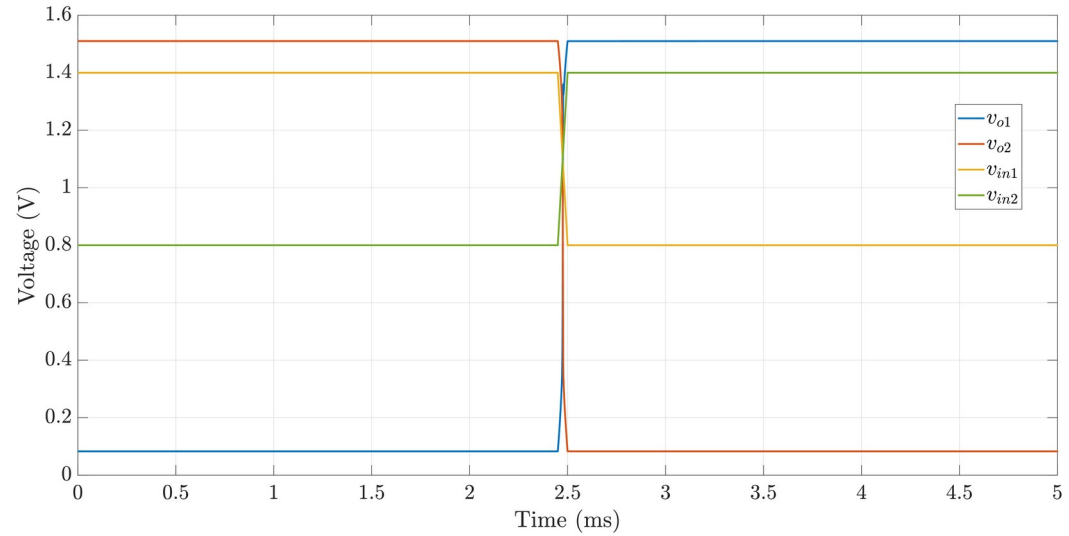
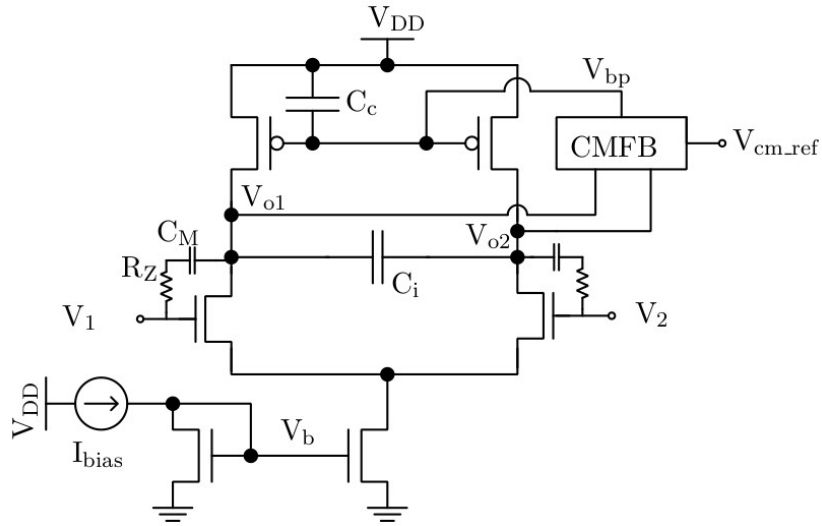
Layout dimensions : 1.2mm x 1.12mm

Designed in **Skywater 130nm**

Supply voltage : 1.8V



Post layout simulation results : Integral controller



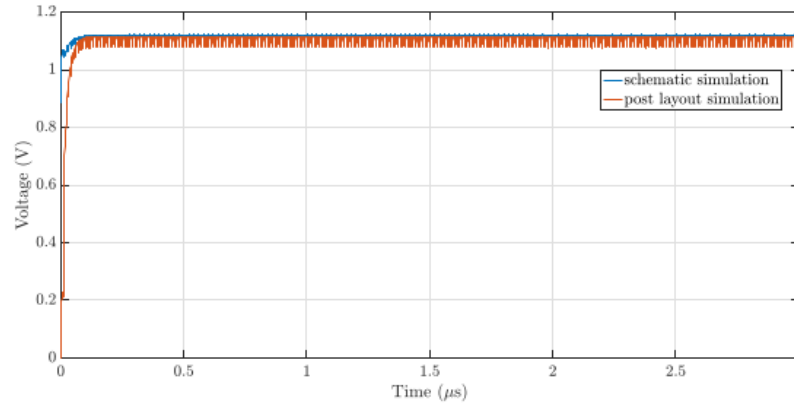
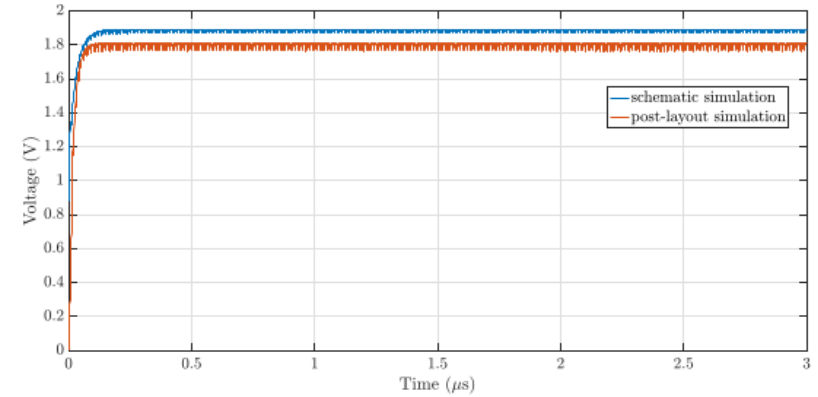
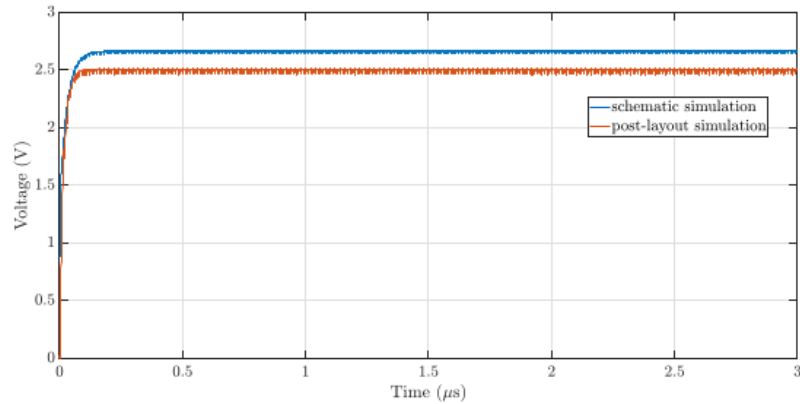
Post layout simulation of controller block

v_{in1} and v_{in2} are input signals.

v_{out1} and v_{out2} are output signals.

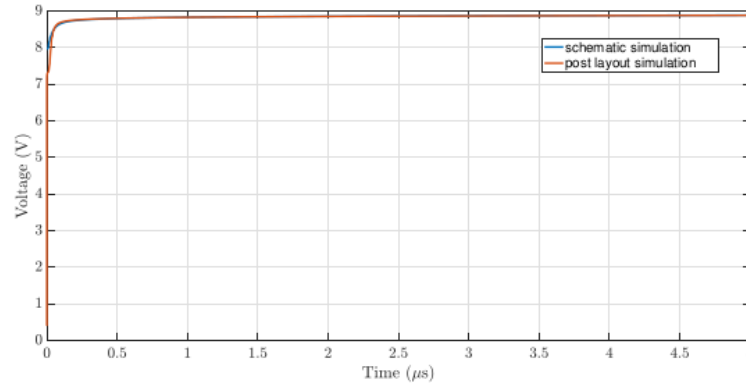
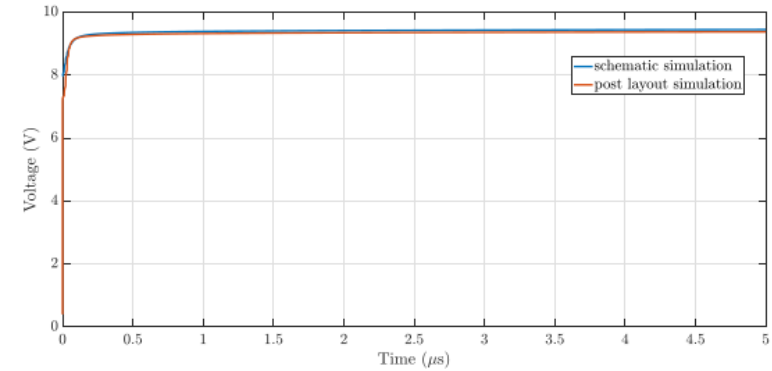
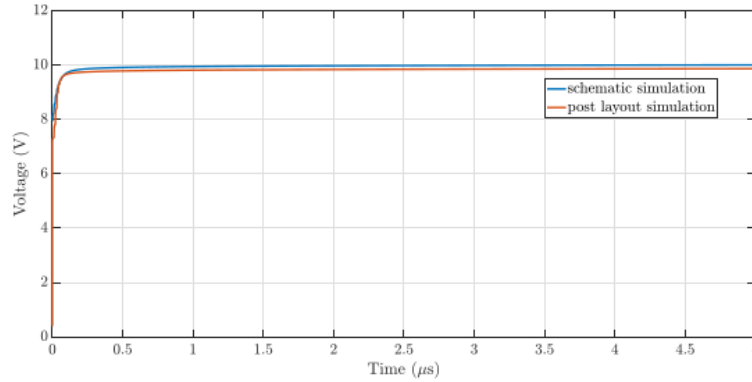
Output saturates due to high gain.

Post layout simulation results : Charge pumps



Post-layout simulation results of charge pump CP-I at highest, common-mode and least efficiency respectively for an input voltage of 0.9 V.

Post layout simulation results : Charge pumps



Post-layout simulation results of charge pump CP-II at highest, common-mode and least efficiency respectively for an input voltage of 9 V.

Conclusions

- Discusses SGFET based sensors as better alternative to passive capacitive accelerometers.
- Discusses and presents simulated results of interfacing circuits.
- Proposes and presents simulated results of SGFET based closed-loop accelerometer for an input range of $\pm 4g$ at frequency of 100Hz designed in UMC180nm.
- Presents post layout simulated results of SGFET based accelerometer designed in SkyWater130nm.

Future work

- Better models for SGFET which includes noise performance.
- Closed loop simulation with more accurate models for device.
- Design to be tested and performance should be verified.
- Design of P-I-N diode based charge pumps to further extend the maximum output voltage that can be generated and design to optimize latency.
- Quantitative analysis of temperature dependence of the various blocks of the system.

Inventory's components

Inventory is a collection of items that are held in stock for the purpose of meeting future demand. It can be used to smooth production, to protect against uncertainty in demand, or to take advantage of bulk discounts.



System's response

When the system is subjected to a step change in the input, the output will change. The system's response to a step change in the input is called the **step response**. The step response is a plot of the output versus time. The step response is used to determine the system's stability, transient response, and steady-state response.



Block diagram of a control system

Exercise 1

1. The following table shows the number of people who attended the concert on each day.

TABLE 1: Number of people attending the concert on each day

Acknowledgements

I would like thank Pramod Martha and Prof. Seena V(from IIST) for their inputs. The simulations to generate values in the LUTs were performed at their group at IIST.

I would like to thank Prof. Abhijit K for his inputs in understanding multi-level converter circuits.

I would like to thank and mention Pranav(final year B.Tech student at IITDh) for his involvement and contribution in tapeout of our design.

I would like to thank Rudra Ramu, MS student MMAE department, for his time and help in SolidWorks diagrams.